

EEE554 – Probability and Random Processes

Module 6: Conditional Distributions

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From Conditional Probability to Conditional Distributions

We have seen that the distribution of a random variable X is described by

$$F_X(x) = P(X \leq x).$$

We have also seen that new information changes probabilities through the conditional probability

$$P(A|M).$$

Question:

How does new information change the distribution of X ?

Suppose we learn that an event M has occurred.

The probability

$$P(X \leq x)$$

is updated to

$$P(X \leq x|M).$$

This naturally leads to the notion of a *conditional distribution*.

Conditional Distributions

Recall the definition of conditional probability.

For events $A, M \in \mathcal{F}$, $P(M) > 0$,

$$P(A|M) = \frac{P(A \cap M)}{P(M)}.$$

Suppose now that $A = \{X \leq x\}$, for a random variable X . Then

$$P(A|M) = P(X \leq x | M),$$

which motivates the following definition.

$$P(X \leq x | M) = \frac{P(X \leq x \cap M)}{P(M)}$$

Quiz

Conditional CDF

Definition 1. *The conditional cumulative distribution function of X given an event M is*

$$F_{X|M}(x) = P(X \leq x \mid M).$$

Equivalently,

$$F_{X|M}(x) = \frac{P(\{X \leq x\} \cap M)}{P(M)}.$$

Conditional PDF

If X is continuous, the conditional probability density function is

$$f_{X|M}(x) = \frac{d}{dx} F_{X|M}(x).$$

Since $P(\cdot|M)$ is itself a probability measure, $F_{X|M}(x)$ is a valid CDF and $f_{X|M}(x)$ is a valid PDF.

Example

Suppose $M = \{X \leq a\}$. Compute

$$F_{X|X \leq a}(x).$$

Observe that

$$F_{X|X \leq a}(x) = \frac{P(\{X \leq x\} \cap \{X \leq a\})}{P(X \leq a)}.$$

Two cases arise.

If $x \leq a$, then

$$\{X \leq x\} \cap \{X \leq a\} = \{X \leq x\},$$

so

$$F_{X|X \leq a}(x) = \frac{F_X(x)}{F_X(a)}.$$

If $x > a$, then

$$\{X \leq x\} \cap \{X \leq a\} = \{X \leq a\},$$

and therefore

$$F_{X|X \leq a}(x) = 1.$$

Combining both cases,

$$F_{X|X \leq a}(x) = \begin{cases} \frac{F_X(x)}{F_X(a)}, & x \leq a, \\ 1, & x > a. \end{cases}$$

Differentiating,

$$f_{X|X \leq a}(x) = \begin{cases} \frac{f_X(x)}{F_X(a)}, & x \leq a, \\ 0, & x > a. \end{cases}$$

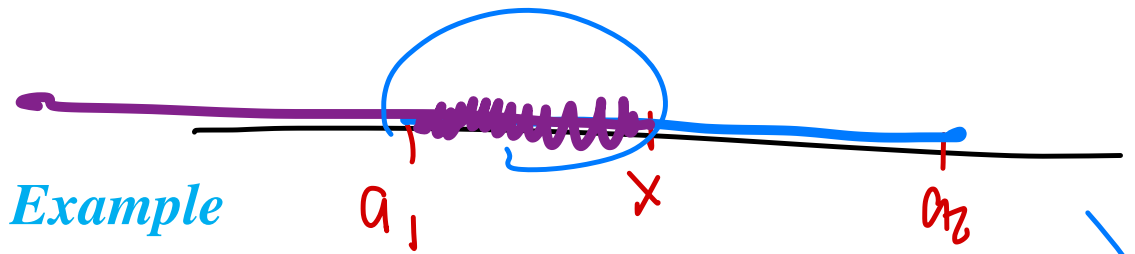
Interpretation

Conditioning on

$$X \leq a$$

removes all probability density above a and renormalizes the remaining density so that its integral is one.

Example



Suppose $M = \{a_1 < X \leq a_2\}$. Find

$$F_{X|M}(x) =$$

$$\frac{P(a_1 < X \leq a_2 \cap X \leq x)}{P(a_1 < X \leq a_2)}$$

Three cases arise.

$$F_{X|M}(x) = \begin{cases} 0, & x \leq a_1, \\ \frac{F_X(x) - F_X(a_1)}{F_X(a_2) - F_X(a_1)}, & a_1 < x \leq a_2, \\ 1, & x > a_2. \end{cases}$$

$$\frac{P(a_1 < X \leq x)}{P(a_1 < X \leq a_2)} = \frac{F(x) - F(a_1)}{F(a_2) - F(a_1)}$$

Hence,

$$f_{X|M}(x) = \begin{cases} \frac{f_X(x)}{F_X(a_2) - F_X(a_1)}, & a_1 < x \leq a_2, \\ 0, & \text{otherwise.} \end{cases}$$

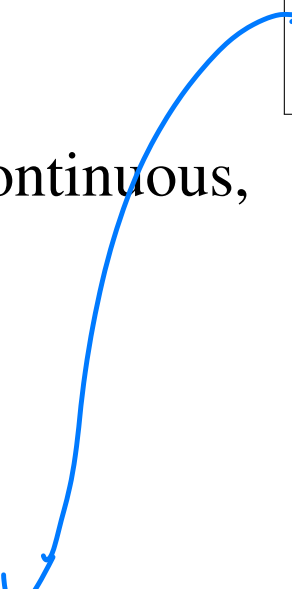
Law of Total Probability

Suppose $\{A_1, \dots, A_n\}$ is a partition of Ω . Then

$$F_X(x) = \sum_{i=1}^n F_{X|A_i}(x) P(A_i).$$

If X is continuous,

$$f_X(x) = \sum_{i=1}^n f_{X|A_i}(x) P(A_i).$$


$$\begin{aligned} P(X \leq x) &= \sum_{i=1}^n P(X \leq x \cap A_i) \\ &= \sum_{i=1}^n P(X \leq x | A_i) \cdot P(A_i) \end{aligned}$$

Conditioning on a Continuous Observation

Recall that for a continuous random variable,

$$P(X = x) = 0.$$

So what do we mean by

$$P(A_i \mid X = x)?$$

We can interpret it by conditioning on a small interval around x :

$\lim_{\Delta \rightarrow 0}$

$$P(A_i \mid x - \Delta/2 < X \leq x + \Delta/2).$$

Using Bayes' rule,

$$P(A_i \mid x - \Delta/2 < X \leq x + \Delta/2) = \frac{P(x - \Delta/2 < X \leq x + \Delta/2 \mid A_i)P(A_i)}{P(x - \Delta/2 < X \leq x + \Delta/2)}.$$

For small Δ ,

$$P(x - \Delta/2 < X \leq x + \Delta/2) \approx f_X(x)\Delta,$$

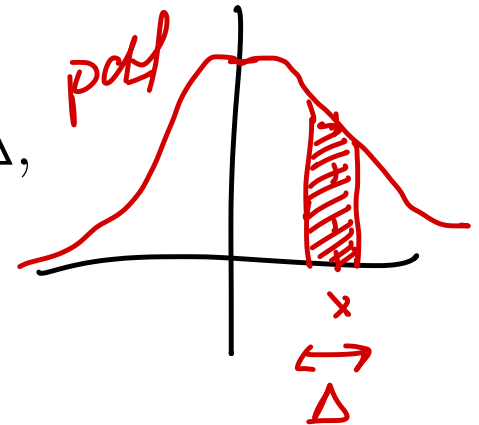
and

$$P(x - \Delta/2 < X \leq x + \Delta/2 \mid A_i) \simeq f_{X|A_i}(x) \cdot \Delta$$

Therefore,

$$P(A_i \mid x - \Delta/2 < X \leq x + \Delta/2) \approx \frac{f_{X|A_i}(x)P(A_i)}{f_X(x)}.$$

$$f_{X|A_i}(x) \cdot \Delta$$



Bayes for events A,B

Taking the limit as $\Delta \rightarrow 0$, gives:

Bayes' Rule

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Suppose X is continuous. Using Bayes' theorem,

$$P(A_i|X = x) = \frac{f_{X|A_i}(x)P(A_i)}{f_X(x)}.$$

Although the event $X = x$ has probability zero, observing a continuous value x can still provide information about which event A_i occurred. The conditional probability is defined through the local behavior of the probability densities around x .

Interpretation

$f_{X|A_i}(x)$ Likelihood: how plausible the observation x is under A_i

$P(A_i)$ Prior: belief in A_i before observing X

$f_X(x)$ Marginal density of X at x (normalizing factor)

$P(A_i|X = x)$ Posterior: belief in A_i after observing x